



Comment on Latinović et al. “Beyond Waste Valorization: Glycerol-Based Metalworking Fluids as Hephaestus for the Circular Economy and Sustainability Transitions”

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Abstract: For many years, we have been working on a new, sustainable lubricant concept based on mixtures of glycerol, water and performance additives. This is a product family with a growing range of applications. Characteristic features include the fact that these lubricants are free of mineral oil and biocides, are based on renewable raw materials and have a high technological performance. The topic of the use of glycerol in lubricants was recently taken up by Latinović et al. and used for a comprehensive, theoretical consideration concerning circular economy and sustainability transitions. The approach seems very interesting. Nevertheless, a number of discrepancies were identified between the theoretical interpretations and the practical experience with glycerol/water-based lubricants. This commentary makes a contribution to naming inconsistencies and bringing theory and practice closer together by an application-oriented discussion. The topics include the description of the practical procedure in lubricant research and development (R&D), the possibilities of using glycerol in novel lubricants, the control of the release of harmful substances from glycerol-containing lubricants, the attempt to predict application limits of these novel lubricants, the aspects to be considered when calculating costs of these lubricants and the interaction of their market launch and application with regulatory and political requirements.

Keywords: Glycerol; Cooling lubricant; Hydraulic fluid; Glycerol-based lubricant; Renewable resources; Decarbonization

1. Introduction

In Germany, for example, metals are machined and formed in around 200,000 companies. Cooling lubricants are used to cool, lubricate, rinse off chips, improve surface quality and tool life and guarantee process reliability. They represent a significant cost, resource and environmental factor. Around 80,000 tons of mineral oil are used in these cooling lubricants every year (Klieber, 2026). In the interests of sustainable industrial production, mineral oil should be replaced by products made from renewable raw materials in order to contribute to the decarbonization of the economy. The Institute for Sustainable Chemistry (ISC) is working on this. Initially, ester-based cooling lubricants derived from renewable raw materials were developed. In this context, the ISC conceived the idea of using a mixture of glycerol and water as a base fluid for lubricants. Following preparatory work, the first collaborative research and development (R&D) project on this topic was launched in 2008, and the ISC has been continuously working on it ever since. It is very gratifying that this novel lubricant concept is now being considered in overarching review articles, such as those by Di Martino (2021), Latinović et al. (2026a), and Latinović et al. (2026b), for which the authors are to be thanked. However, upon reading these articles, it becomes clear that they are theoretical studies in which far-reaching conclusions are sometimes drawn exclusively from the literature. In our view, this calls for commentary, supplementation, and corrective measures from the practitioners involved in the R&D of glycerol/water-based lubricants. That is the aim of this article. In doing so, reference is made both to the article mentioned in the title (Latinović et al., 2026a) and to the thematically closely related article (Latinović et al., 2026b), published at the same time.

2. Methodology

In this section the strategy and approach of the ISC and its partners regarding R&D work on glycerol/water-based lubricants is outlined. Initially, pure glycerol/water mixtures were prepared and analyzed. Of particular interest were the increasing corrosion protection with rising glycerol content, suitable viscosities, and complete protection of the mixture against microbial growth starting at a glycerol content of 30% by mass. When diluted below 30%, the glycerol is highly biodegradable. In addition, experiments on the surface machining of steel (100Cr6) by grinding showed that this is already possible, for example, with an additive-free glycerol/water mixture (40:60 by mass), whereby the water provides cooling and the glycerin provides lubrication. Such a machining process was not feasible with pure mineral oil. It is advantageous if the base fluid already exhibits desirable properties for the intended application, which need only be enhanced through the addition of performance additives. The water-soluble polymer carboxymethyl chitosan, which was added to the base fluid at a later stage, also proved to be multifunctional in the specification used here, among other things due to its thickening effect, its ability to establish a basic pH typical of lubricants, and its easy adsorption onto metal surfaces as a polyanion.

It was decided to use this base fluid to develop novel lubricants. It was specified that all components should always be mutually soluble in a polar manner. The formation of emulsions should be avoided. The water content should always be at least 35%, which means that the fluid variants are considered as hardly inflammable (Deutsches Institut für Normung e.V., 2020). The glycerol should also be present at $\geq 35\%$ to reliably ensure protection against microbial contamination without having to use biocidal additives. The fluid variants should consist entirely of renewable raw materials, be environmentally friendly and biodegradable, exhibit very low human toxicological potential, and demonstrate high technological performance.

The R&D projects are always carried out in collaboration with partners. The Institute of Machine Tools and Manufacturing Technology at the Technical University of Braunschweig (TU Braunschweig) has a range of modern, industrial-grade metalworking machines for testing cooling lubricants under conditions that closely resemble actual production. At the Institute for Mobile Machinery and Commercial Vehicles at TU Braunschweig, complex hydraulic systems are set up for large-scale, realistic testing of hydraulic fluids. In addition, industry partners from the fields of chemistry (lubricant manufacturers) and engineering (component and plant manufacturers) are involved in each project. Our research group focuses on chemical analysis. All components of a lubricant, as well as contaminants and degradation products, can be measured both qualitatively and quantitatively (Stache et al., 2013). An R&D project typically consists of the following phases: (a) definition of chemical and technological requirements, (b) chemical, physical, and tribological investigations in the laboratory, (c) practical tests in pilot plants, and (d) assessment of environmental and human safety. In all cases, a commercially available lubricant designed for the same application is tested in parallel. The novel lubricant should demonstrate comparable or better technical performance. The overarching goal is to develop a growing product range based on this lubricant concept.

3. Results and Discussion

This section addresses aspects from the two review articles (Latinović et al., 2026a; Latinović et al., 2026b) and comments on, supplements, or critically discusses them from the perspective of R&D practice.

3.1 The Use of Glycerol in Lubricants

It has been sufficiently described that glycerol is produced in large quantities as a by-product in the production of biodiesel and other fatty acid esters and has not yet been fully utilized as a material. New utilization options are being sought, such as the use in lubricants. The term “circular economy” is somewhat vague in this context. A definition of the European Parliamentary Research Service (EPRS) states: “Circular economy is based on sharing, leasing, reuse, repair, refurbishment and recycling, in an (almost) closed loop, where products and the materials they contain are highly valued” (Bourguignon, 2016). To put it more precisely, in connection with the novel cooling lubricant concept one should rather speak of cascading use that should be as multi-stage as possible. Here is an example scenario comprising a six-stage use, starting with vegetable oil as a renewable raw material and ending up with the conversion of the final organic chemical methane into thermal energy:

1. Cooking fat is used to fry food.
2. After its service life, the cooking oil is collected (currently approx. 400,000 t/a in Germany) and used to produce biodiesel.
3. The by-product glycerol could be technically applied in lubricants.
4. Some of the fluid resulting from water-based washing processes of workpieces and swarf contaminated with lubricant could be recycled after purification, i.e. fed back into the lubricant in use.
5. After the end of its application (1.0–1.5 years), the lubricant could serve as a substrate in biogas plants. It is known that glycerol has a positive effect as a booster in the biogas production process (Fountoulakis et al.,

2010).

6. The resulting methane would ultimately be converted into energy by combustion.

As can be seen from the example, it is only possible to recycle part of the cooling lubricant (step 4). For the rest, the material use of the renewable raw material and the intermediates progresses until, in the end, only energy recovery of the final carbon compound remains (typical cascading use).

The question of glycerol qualities suitable for lubricants has been clarified from the authors' point of view. We are not aware of any technological application in which pure glycerol is used as a lubricant. Water is always added for dilution in order to adjust the viscosity of the mixture. Crude glycerol usually contains solid particles, organic substances, salts, polymers as well as colorants and odorants. It is therefore unsuitable for complex technological applications. Ultrapure or pharmaceutical grade glycerol ($\geq 99\%$) is too expensive. Between these two extremes, glycerol grades are offered with various designations (e.g. 85% Ph. Eur., 87%, technical), which mostly contain $\leq 15\%$ water and contain other impurities, mostly salts, only to an insignificant extent. They are around one-third cheaper than pharmaceutical glycerol. Only the water content should be determined before using them. Less water should then be added to the lubricant formulation accordingly.

3.2 Release of Hazardous Substances Due to Glycerol Decomposition

When the development work on glycerol-containing lubricants became known, the question immediately arose as to whether the glycerol decomposes under thermal stress during use and whether harmful substances, such as acrolein, are released as a result. The thermal decomposition process of pure glycerol is described in the literature. This problem was taken very seriously and was analyzed in detail in every R&D project. For example, worst-case tests were carried out in the laboratory with glycerol-containing fluid variants, which included thermal loads in contact with various metal surfaces and changing pH values. Samples of both the fluids and the gas space above the fluids under test were analyzed. In the technical center, gas samples were taken both at the processing machines in the working area of the operating personnel and at the air extraction system, as well as in the machines, e.g. 30 cm away from the grinding wheel in operation. Various samplers were used so that a wide range of substances from polar to non-polar could be recorded. For aldehydes and ketones, which include acrolein, a special, very sensitive analytical method was applied. The samples were analyzed using gas chromatography coupled with mass spectrometry (GC/MS) and high-performance liquid chromatography (HPLC) with high detection sensitivities. In fact, there were no worrying findings in any case (Winter et al., 2012).

Acrolein was specifically mentioned in the reviews (Latinović et al., 2026a; Latinović et al., 2026b) and highlighted as a hazard. This substance currently has a European occupational exposure limit value of 0.05 mg/m³ of breathing air. The detection limit of our measurement technology is 0.005 mg acrolein/m³ air. Nevertheless, we could not find any indication of this substance in any case. Is this plausible from a chemical point of view? Grinding steel is the machining process we have investigated to date in which the new type of cooling lubricant is subjected to the greatest stress. In test operation, the machine's tank was filled with 150 L of cooling lubricant, which was circulated. As usual, the so-called flooding lubrication was used, whereby the workpiece and the tool, and especially the contact point between the two, are continuously supplied with a large amount of cooling lubricant. A small hot spot is created during the grinding process, with which a small amount of the cooling lubricant comes into contact. From the point of view of reaction kinetics, the glycerol is heated for a very short time and immediately cooled back down to room temperature, whereby the water with its high heat capacity absorbs a large proportion of the heat. From this point of view, the time at high temperature is too short for complex decomposition reactions to take place. One product of the decomposition reactions would be water, which is already present as a main component in the cooling lubricant. From the point of view of thermodynamics (here: Le Châtelier's principle of least constraint), the decomposition reactions are pushed in the direction of the educt, i.e. glycerol, by the excess water, which also works against its decomposition. The non-occurrence of thermally induced decomposition products of glycerol appears plausible under these assumptions. Thermal instability should not be emphasized as a disadvantage of glycerol-based lubricants as long as this has not been proven in practical tests.

3.3 Prediction of the Application Limits for Glycerol-Based Lubricants

There is a wide range of applications for lubricants with very different requirements. The use of performance additives offers various possibilities for changing and adapting properties and for a "fine-tuning" of the lubricant for the respective application. A large number of non-polar soluble additives are available for lubricants containing mineral oil. Substances such as zinc dialkyldithiophosphates have been used in some cases since the middle of the 20th century (Wichmann et al., 2010). The selection of polar-soluble additives however is comparatively small. A considerable part of the research work of our working group therefore consists of developing new polar-soluble additives. In addition, we are currently working on modifying even the base fluid and, for example, replacing glycerol with other substances that can be produced from renewable raw materials (Wiesbaum et al., 2025). So far, lubricant variants for grinding steel (Wichmann et al., 2013; Winter et al., 2012), for machining and shaping

aluminium alloys (Leiden et al., 2023) and for hydraulic fluids (Gelinski et al., 2016) have been successfully developed according to the concept discussed here (Figure 1). However, there are still many options for opening up further fields of application for this sustainable lubricant concept; it is certainly not yet possible to set limits here. A consideration of the pure base fluid consisting of glycerol and water is certainly not expedient for estimating the application limits.

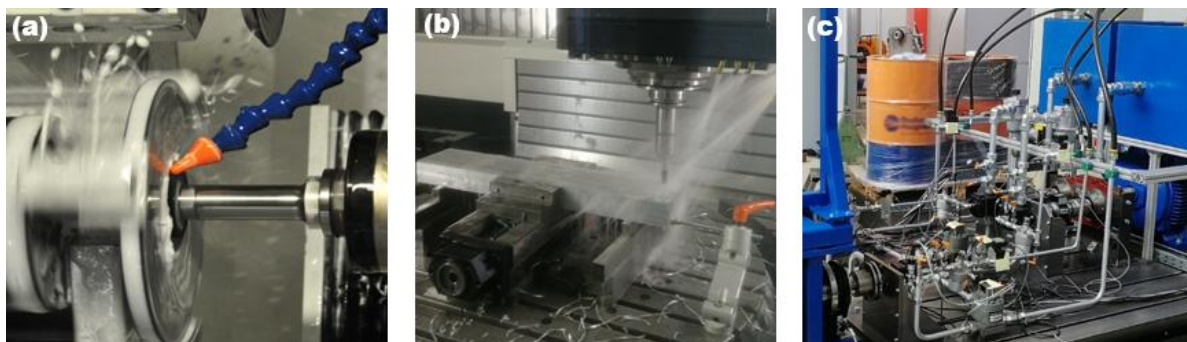


Figure 1. Applications of glycerol/water-based lubricant variants: (a) steel machining; (b) aluminium machining; and (c) hydraulic fluid testing at full technological scale

3.4 Cost Calculations for the Use of the New Lubricants

A cooling lubricant based on renewable raw materials rather than mineral oil does not automatically have a good chance of a successful market launch simply because it is considered a sustainable product. Rather, it is in direct competition with conventional cooling lubricants, with cost and technical performance being the decisive factors. These aspects are thus directly linked to the topic of sustainable business practices and must therefore be considered here, just as they are in the publications (Latinović et al., 2026a; Latinović et al., 2026b). Aspects of technical performance are especially addressed in sections 3.1, 3.3, and 3.5, including Figure 1, while the cost aspect is discussed in this section below.

Previous calculations and estimates have resulted in production costs for variants of glycerol/water-based fluids of between 1.50 €/kg and 6.00 €/kg. This means that they are already competitive with many of the mineral oil-based commercial products in terms of production costs, which is being further increased by the price explosion for mineral oil caused by current crises and wars. However, the purchase price of the lubricant should not be considered in isolation. Rather, lubricant properties drive various downstream costs incurred during production and lubricant application, all of which must be factored into total operational cost assessments.

Once again, the surface treatment of steel by grinding is taken as an example. First of all, the tools, in this case the grinding wheels, are expensive and the cooling lubricant has a major influence on the wear and thus the replacement interval of these tools. The high tribological performance of the new cooling lubricant will contribute to comparative cost savings here. The workpiece surfaces must be cleaned after production and before further processing in order to be suitable for bonding or painting, for example. Today, this is often done conventionally using washing baths containing aqueous surfactant solutions. The disposal of washing solutions containing mineral oil, which are produced in large quantities in industry, causes costs in the region of 300 €/t. The novel, polar-soluble cooling lubricant can be cleaned by rinsing with water, whereby the rinsing solution is potentially recyclable. This simplified procedure should contribute to a further significant reduction in costs. In Germany, an estimated 300,000 tons of metal chips are produced each year from metalworking processes. Usually, they are wetted with cooling lubricant, in some cases the term grinding sludge applies. If the swarf contains more than 8% organics, it is classified as hazardous waste, which is associated with high disposal costs. If the organic content is lower, however, the metal swarf can be sold at a profit to steelworks, foundries or cement plants. However, chip de-oiling is technically demanding and therefore expensive if they are wetted with mineral oil. In a simple cleaning experiment with chips wetted with a glycerol/water-based cooling lubricant, 892 mg of organically bound carbon could be dissolved from 4,588 mg of chips with the aid of 200 mL of water (Wichmann et al., 2013). Here again, high reductions in process costs can be expected compared to conventional mineral oil-based cooling lubricants. Advantageous options for recycling the novel sustainable fluids at the end of their service life have already been described in section 3.1.

3.5 Regulation, Standardization and Political Requirements

The question of market introduction and application regulation of novel sustainable lubricants needs to be discussed. A first answer is to research and develop a growing product family for ever more extensive applications

in order to avoid the creation of a good but unnoticed niche product. Even a grinding machine for steel processing, for example, is operated with different lubricants, e.g. spindle oil, bed carriage oil, hydraulic oil, and cooling lubricant, which come into contact with each other and must be compatible. Another aspect that is not directly monetary is that environmentally friendly and human-compatible products are increasingly being advertised, sold and used today. A current project is therefore working on determining whether it would be possible to issue an environmental certificate such as the German Blue Angel (Umweltzeichen Blauer Engel, 2026) for a representative of the new product range. A life cycle assessment is also being prepared for the lubricant.

Lubricant manufacturers and users often know each other. In our opinion, they tend to act conservatively when it comes to introducing fundamental innovations (“Never touch a running system”). If a competitor makes an advance, successes and failures are closely monitored by the community. In the further development of the innovative lubricant concept, care is always taken to ensure that the technical performance of the fluid variants is ultimately comparable to or better than that of conventional, commercially available lubricants for the corresponding application. In addition, the new lubricants are designed in such a way that they can, in principle, be used in existing machines without major technical modifications, i.e. they are largely compatible with existing industrial infrastructure. Keywords here include pump and filter compatibility, viscosity adjustment, material compatibility as well as technical and occupational safety. Of course, fine-tuning for a specific application would always have to be carried out by the manufacturers and users of the lubricant. The task of R&D in the project network is merely to develop fluids that function well in systems that are as close to reality as possible.

If the conditions described above are met, cost accounting will ultimately remain the decisive driver for a switch to sustainable lubricants. Rising mineral oil prices and the vulnerability of the lubricant sector’s raw material supply to crises will promote innovation. Political guidelines as well as regulatory and standardization measures may be helpful at a later stage, but could hinder or stifle the introduction of innovations of this kind at an early stage.

To summarize, the following approach might be, from our point of view, the most promising for the market launch of innovative, sustainable metalworking fluids:

- R&D towards a product family based on a sustainable cooling lubricant concept; proof in principle of high technological performance.
- Fine-tuning of products for defined applications by coolant manufacturers in cooperation with industrial users. Ensuring compatibility with existing technologies and legal regulations. Financial competitiveness in terms of overall costs.
- Demonstration and documentation of practical suitability in industrial applications; in parallel, expansion and improvement of the product range.
- Subsequently, as for example with the introduction of electromobility, political support in terms of promoting a sustainable economy and lifestyle as well as standardization and regulatory measures could follow, if necessary.

4. Conclusion

The overarching view of a concept such as the development of glycerol-based lubricants as a contribution to the decarbonization of the economy is to be welcomed. Nevertheless, a comparison with practice is necessary in order to avoid misleading interpretations and conclusions. The R&D work to date has benefited greatly from the collaboration between various scientific disciplines and industrial sectors. An expansion of interdisciplinary cooperation also in this direction is possible at any time and it appears advantageous.

In terms of content, it will continue to be important to develop new polar-soluble performance additives and alternative components for the base fluid in order to improve the performance of the innovative, sustainable lubricants and to expand their range of applications. The preparation of the issuance of environmental certificates (ecolabels) and the development of life cycle assessments represent a further level of consideration for these sustainable lubricants.

Author Contributions

Conceptualization, H.W., M.K., and J.G.; validation, H.W., M.K., and J.G.; writing—original draft preparation, H.W.; writing—review and editing, H.W., M.K., and J.G.; visualization, H.W. All authors have read and agreed to the published version of the manuscript.

Data Availability

Additional information on this comment is available from the corresponding author upon reasonable request.

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Conflicts of Interest

The authors declare no conflicts of interest.

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